LAB 03 - BASIC ANALYSIS

MALWARE ANALYSIS AND INCIDENT FORENSICS
M.Sc. in Cyber Security

MALWARE ANALYSIS
M.Sc. in Engineering in Computer Science



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CYBER INTELLIGENCE AND INFORMATION SECURITY

LAB_03-1.EXE

Answer the following questions:

1) When was this file compiled?
2) List a few imports or sets of imports and describe how the malware might use them.
3) What are a few strings that stick out to you and why?
4) What happens when you run this malware? Is it what you expected and why?
5) Name a procmon filter and why you used it.
6) Are there any host-based signatures? (Files, registry keys, processes or services, etc). If so, what are they?
7) Are there any network based signatures? (URLs, packet contents. etc) If so, what are they?
8) Is there anything that impeded your analysis? How so? How might you overcome this?
9) What do you think is the purpose of this malware?

LAB_03-1.EXE

Hints:

- Use a software that allows you to inspect the PE Header
  - Open the EXE in PeStudio
- Check the “compiler-stamp” value for the compilation date
- Look at the “Imports” tab
  - ShellExecuteExA - launches other processes via the shell
- Look at the “strings” tab
  - Notice “60.248.52.95:443” !!!

LAB_03-1.EXE

Hints:

- Use a software to monitor the malware execution
  - e.g. ProcMon
  - Analyze the trace looking for actions
- Filter out everything that is not directly or indirectly related to the EXE
- Look for network activities
- Look for file-level activities

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LAB_03-2.EXE

Answer the following questions:

1) What is the md5sum? What of interest does VirusTotal Report?
2) List a few imports or sets of imports and describe how the malware might use them.
3) What are a few strings that stick out to you and why?
4) What happens when you run this malware? Is it what you expected and why?
5) Name a procmon filter and why you used it.
6) Are there any host-based signatures? (Files, registry keys, processes or services, etc). If so, what are they?
7) Are there any network based signatures? (URLs, packet contents. etc) If so, what are they?
8) Is there anything that impeded your analysis? How so? How might you overcome this?
9) What do you think is the purpose of this malware?

LAB_03-3.EXE

Answer the following questions:

1) Are there any indications that this malware is packed? What are they? What is it packed with?
2) Are you able to unpack it? Why or why not?
3) What are a few strings that stick out to you and why?
4) What happens when you run this malware? Is it what you expected and why?
5) Are there any host-based signatures? (Files, registry keys, processes or services, etc). If so, what are they?
6) Are there any network based signatures? (URLs, packet contents. etc) If so, what are they?
7) Is there anything that impeded your analysis? How so? How might you overcome this?
8) What do you think is the purpose of this malware?

LAB_03-4.EXE

Answer the following questions:

1) Are there any indications that this malware is packed? What are they? What is it packed with?
2) Are you able to unpack it? Why or why not?
3) What are a few strings that stick out to you and why?
4) What happens when you run this malware? Is it what you expected and why?
5) Are there any host-based signatures? (Files, registry keys, processes or services, etc). If so, what are they?
6) Are there any network based signatures? (URLs, packet contents. etc) If so, what are they?
7) Is the malware performing actions to persist in the target systems? Which strategy does it use?